Assignment

An assignment for class **MET CS 579** at Boston University, instructed by **Jae Lee**. Student Jin Yu Zhang submitted on **January 20, 2023**.

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I finished the installation of MySQL on my MacBook laptop with M1 chip. Here is the machine information:



MacBook Pro

16-inch, 2021

Chip	Apple M1 Pro
Memory	32 GB
Startup disk	Macintosh HD
Serial number	C19F2N5X4V
macOS	13.1 (22C65)

More Info...

Regulatory Certification

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The `.dmg` files sources:

- **MySQL**: 8.0.32 macOS-arm64
- **MySQL Workbench**: 8.0.31 macOS-x86
  > (`8.0.32` crashed every time I quired)

After the installation finished, I found out that `mysql` wasn't added automatically. My terminal complained `zsh: command not found` when I tried to use `mysql`, and I also couldn't find MySQL in my OS's application list.

> The root password was set up during the installation process

So I added the MySQL path to `~/.zshrc` in order to use `mysql` with terminals without entering the full path:

```
# ~/.zshrc
...
# >>> mysql setup >>>
export PATH=/usr/local/mysql/bin:$PATH
# <<< mysql setup <<<
```

Then ran the following command to activate the script:

```
source ~/.zshrc
```

Users

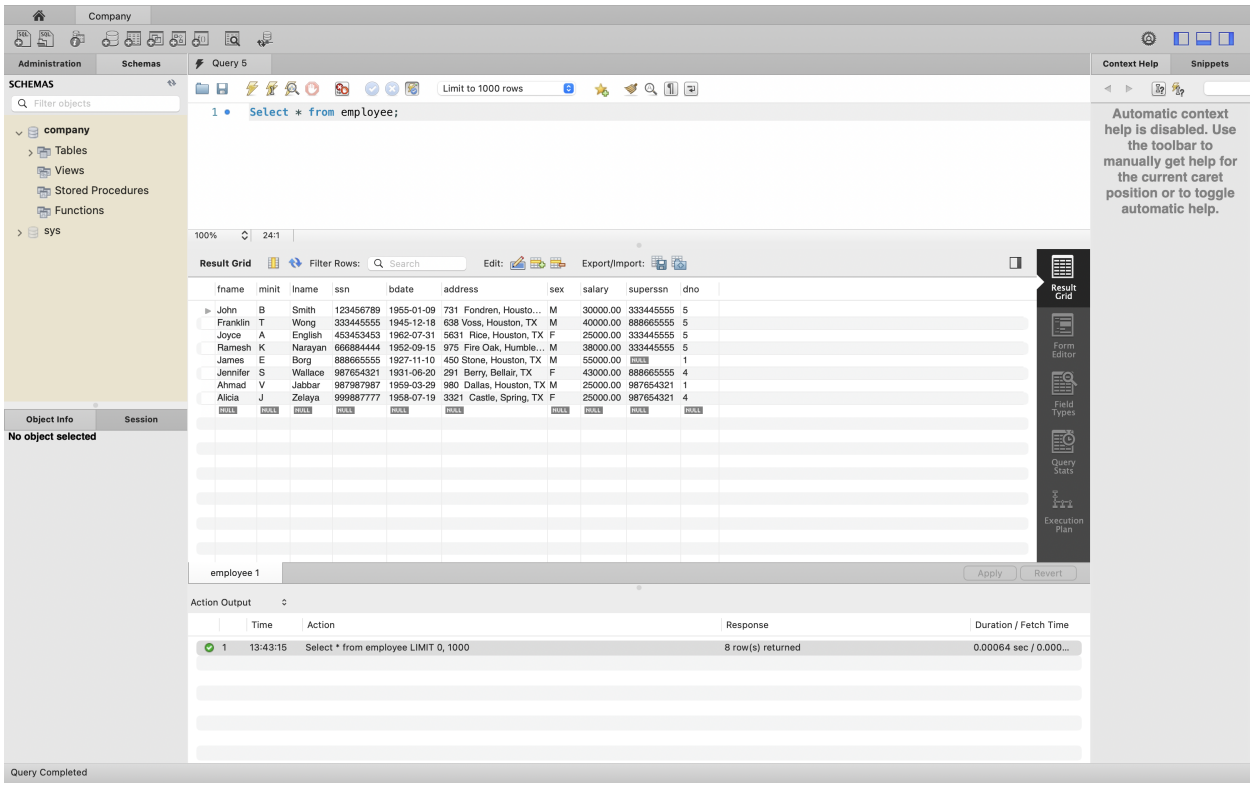
I then used `mysql --user root --password` to login, creating users with the following SQL. Passwords are not hidden since I only allow the users to connect from `localhost`:

```
CREATE USER 'company'@'localhost' IDENTIFIED WITH caching_sha2_password


GRANT ALL PRIVILEGES ON *.* TO 'company'@'localhost' WITH GRANT OPTION;
```

Query

After populating data to the local database with `company_tables.sql` and `company_tuples.sql`, I could successfully use `SELECT` to query:




```
(base) lookswalidy@Zhangs-MBP ~ % mysql --user company --password
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 121
Server version: 8.0.32 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| company            |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.00 sec)

mysql> USE company;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+------------------+
| Tables_in_company |
+------------------+
| department       |
| dependent        |
| dept_locations   |
| employee         |
| project          |
| works_on         |
+------------------+
6 rows in set (0.00 sec)

mysql> SELECT * FROM employee;
+----------+-------+---------+-----------+------------+--------------------------+------+----------+-----------+------+
| fname    | minit | lname   | ssn       | bdate      | address                  | sex  | salary   | superssn  | dno  |
+----------+-------+---------+-----------+------------+--------------------------+------+----------+-----------+------+
| John     | B     | Smith   | 123456789 | 1955-01-09 | 731
Fondren, Houston, TX | M    | 30000.00 | 333445555 |    5 |
| Franklin | T     | Wong    | 333445555 | 1945-12-18 | 638 Voss, Houston, TX    | M    | 40000.00 | 888665555 |    5 |
| Joyce    | A     | English | 453453453 | 1962-07-31 | 5631
Rice, Houston, TX   | F    | 25000.00 | 333445555 |    5 |
| Ramesh   | K     | Narayan | 666884444 | 1952-09-15 | 975
Fire Oak, Humble, TX | M    | 38000.00 | 333445555 |    5 |
| James    | E     | Borg    | 888665555 | 1927-11-10 | 450 Stone, Houston, TX   | M    | 55000.00 | NULL      |    1 |
| Jennifer | S     | Wallace | 987654321 | 1931-06-20 | 291
```